UML Class Diagram Specification Report

This document provides a detailed breakdown of the classes, attributes, and methods identified in the provided UML class diagram for the University/Institution Management System.

1. Core User & Role Entities

User

- **Attributes:** id (Serial PK), name (Text), email (Text UNIQUE), password_hash (Text), role (Enum: admin/faculty/student), institution (Text), admin_id (Integer FK), department_id (Integer FK), created_at (Timestamp)
- **Methods:** viewTimetable()

Admin (Role)

Inherits from User

- **Methods:** manageDepartments(), manageFaculty(), manageRooms(), setupConstraints(), reviewLeaves(), publishTimetable()

Student (Role)

Inherits from User

- **Attributes:** department_id (Integer FK)
- **Methods:** viewTimetable()

2. Academic & Administrative Entities

Faculty

- **Attributes:** id (Serial PK), user_id (Integer FK UNIQUE), employee_code (Text), max_weekly_hours (Integer), is_present (Integer), admin_id (Integer FK)
- **Methods:** submitLeaveRequest()

Department

- **Attributes:** id (Serial PK), name (Text), level (Text: UG/PG), semester (Integer), section (Text), strength (Integer), admin_id (Integer FK)

Subject

- **Attributes:** id (Serial PK), name (Text), code (Text), semester (Integer), classes_per_week (Integer), admin_id (Integer FK)

Room

- **Attributes:** id (Serial PK), room_id (Text), type (Enum: classroom/lab/hall), capacity (Integer), is_available (Integer), admin_id (Integer FK)

3. Scheduling & Operations

TimetableSlot

- **Attributes:** id (Serial PK), department_id (Integer FK), day (Enum: Mon-Fri), slot_index (Integer), slot_time (Text), subject_id (Integer FK), faculty_id (Integer FK), room_id (Integer FK), is_break (Integer), admin_id (Integer FK)

LeaveRequest

- **Attributes:** id (Serial PK), faculty_id (Integer FK), leave_date (Text), reason (Text), leave_type (Text: casual/medical), status (Enum: pending/approved/denied), admin_note (Text), created_at (Timestamp), reviewed_at (Timestamp)

Substitution

- **Attributes:** id (Serial PK), leave_request_id (Integer FK), timetable_slot_id (Integer FK), original_faculty_id (Integer FK), substitute_faculty_id (Integer FK), department_id (Integer FK), day (Text), slot_index (Integer), subject_id (Integer FK), status (Text), created_at (Timestamp), admin_id (Integer FK)

TimetableHistory

- **Attributes:** id (Serial PK), admin_id (Integer FK), name (Text), generated_at (Timestamp), department_ids (JSONB), snapshot (JSONB), log (Text), status (Text: draft/published), type (Text: regular/substitution), leave_request_id (Integer FK)

Constraint

- **Attributes:** id (Serial PK), name (Text), type (Enum: hard/soft), category (Text), is_enabled (Integer), is_builtin (Integer), config (JSONB), admin_id (Integer FK), created_at (Timestamp)

4. Junction Tables (Relationships)

Table Name	Foreign Keys / Description
faculty_subjects	faculty_id (FK), subject_id (FK)
faculty_departments	faculty_id (FK), department_id (FK)
subject_departments	subject_id (FK), department_id (FK)

Note: All data fields and methods have been verified against the original UML diagram provided.